



**Department of Electrical and Computer
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Lab 6 : “LCD”

Introduction and objectives

Liquid Crystal Displays (LCDs) are widely used in embedded systems to visually convey information such as sensor readings, text messages and data outputs. In this lab, you will learn how our LCD module HD44780U is interfaced with the MC9S12DT256 microcontroller, you will understand the principles behind SPI0, Serial peripheral interface, that is ensuring the communication between MCU and LCD and you will develop, as well, the skills needed to initialize, control the LCD, and display text messages.

Pre Lab6 :

In this preliminary task, you are required to answer the following questions :

1. Locate the LCD on the appropriate schematic diagram and find out how many pins does the LCD_PORT have? Name them all.
2. Look out carefully at the LCD connections, 6 pins of the LCD are connected to the ground through resistors , name these pins.
3. Follow the LCD connections and check that it is connected to the shift register 74HC595.
Name the LCD pins that are connected to 74HC595.
4. Deduce if the LCD is receiving Data from MCU in 8 bits operation or 4 bits operations.

5. Referring to 'HC595' datasheet (located in 'Shift Register' folder), study the 74HC595 :
 - main function
 - Functional diagram (explain how the shifting process is performed).
 - all its pins and their connections with the LCD in the schematic diagram.
6. By following the connections: LCD_Port → 74HC595 → ?
Name the interface connected to 74HC595, and the 3 pins that ensure this connection.
7. By referring to the datasheet S12PSPV2 (located in folder 'SPI'), study the Serial Peripheral Interface SPI0:
 - main function and application
 - block diagram
 - the role of three pins: MOSI, MISO, SCK.
8. Name the 5 registers of SPI0.
9. Study the role of these bits:
 - SPE, MSTR, CPOL, SSOE, LSBFE (*SPI0CR1*).
 - MODFEN (*SPI0CR2*).
 - SPTEF (*SPI0SR*).
10. Study the role of MODRR (Module Routing Register), and deduce the configuration of the bit MODRR4.
(refer to *PBMCUSLK_SCH_D_0* and check the port to which SPI0 was re-routed).

Lab 6 : "LCD"

Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the 2 tasks below:

Task 1:

The first program should do the following:

Initialize the LCD , write the word 'MCO' on the first line , and your group name starting from the second position on the second line as follows: 'G one 'G four'.

Task2 :

The second program should do the following:

Use the manual 'HD44780U_Summary' located in 'Datasheets/LCD' folder

- a) Write the word 'RID' on the first line starting from the first position.
- b) Shift the cursor position to the left 2 times to write the letter 'E' over 'I'.
- c) Write the word 'CAR' on the second line starting from the first position.
- d) Shift the entire display to the left.
- e) Clear the entire display.

In order to initialize and write on the LCD, you have to follow these steps:

Step1: initialize SPI0

Step2: write a time **DELAY** subroutine that generates a time delay ≥ 40 ms (40 or 60 ms for example)

Step3: Send the bytes through SPI0

Step4: Since the LCD is connected to the shift register 74HC595 through 4 data pins: LCD_D4 $\rightarrow$ LCD_D7, it follows the **4 bits operation** mode, we have to send every byte 2 times: first time for the MSB, second time for the LSB.

Step5: prepare the bytes to be sent twice, knowing that there are 2 types of data : **Data** and **instructions**, and that depends on the **RS** bit of the LCD:

If RS=0: we are sending instructions

If RS=1: we are sending Data

Write 2 subroutines: **SENDDATA**: to send Data, and **SENDINST** : to send Instructions.

Step6: By referring to the “HD44780U” datasheet, follow the steps needed to initialize by instruction the 4 bit interface LCD.

Step7: Start sending data bytes or instruction bytes.

Step1 Code: initialize SPI0

- configure the pins of the PORT connected to SPI0 as INPUTS or OUTPUTS (MOSI, SCK, SS :OUTPUT, MISO:INPUT)
- set the corresponding bit of MODRR to choose SPI0
- movb #\$52, SPIOCR1
- movb #\$10, SPIOCR2
- movb #\$00, SPIOBR

Step3 Code: Send the bytes through SPI0

Call this subroutine **SENDSPI**

- staa SPIODR
- brclr SPIOSR,\$20,*
- movb SPIODR,Data ; to reset SPIODR (Data is a variable)
- movb SPIOSR,Data ; to reset SPIOSR
- rts

N.B: WRITE RTS AT THE END OF EACH SUBROUTINE.

Step5 Code: Send instructions byte.

The byte to be sent must have the following pattern:

EN	R/S	*	*	D7	D6	D5	D4
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0

D7 → D4 must have the MSB value first, then the LSB

this is the **SENDINST** subroutine.

- tab ; transfer initial value of 'a' to 'b' to save its value
- rora] ; since the LCD operates in 4 bit mode, we have to send the 4 most
- rora] significant bits first , so I rotate 'a' to the right 4 times .
- rora]
- rora]
- oraa #%10000000] ; LCD requires a Latch (rising edge → falling edge) on the EN ,
- anda #%10001111] ; to capture the data sent , here 'EN' =1
- jsr SENDSPI
- anda #%00001111 ; then 'EN' =0
- jsr SENDSPI
- jsr DELAY
- tba ; transfer the value of 'a' stored in 'b' to start working on the LSB
- oraa #%10000000 ; same as before, we need to make a Latch on pin 'EN'
- anda #%10001111
- jsr SENDSPI
- anda #%00001111
- jsr SENDSPI
- jsr DELAY
- rts

The **SENDDATA** subroutine is the same as **SENDINST** , but you have to change the RS bit to 1.

Step6: Refer to the “HD44780U” datasheet to initialize by instruction the 4 bit Interface LCD.
